

Python String Methods - Detailed Guide

capitalize()

Explanation: Makes the first character uppercase.

Example: `'hello'.capitalize()`

Output: Hello

casefold()

Explanation: Converts to lowercase for case-insensitive comparisons.

Example: `'HeLLo'.casefold()`

Output: hello

center()

Explanation: Centers the string within a given width.

Example: `'hi'.center(6, '-')`

Output: --hi--

count()

Explanation: Counts occurrences of a substring.

Example: `'banana'.count('a')`

Output: 3

encode()

Explanation: Encodes the string into bytes.

Example: `'abc'.encode()`

Output: b'abc'

endswith()

Explanation: Checks whether the string ends with a suffix.

Example: `'file.txt'.endswith('.txt')`

Output: True

expandtabs()

Explanation: Replaces tab characters with spaces.

Example: `'a\tb'.expandtabs(4)`

Output: a b

find()

Explanation: Returns first index of a substring or -1.

Example: `'hello'.find('l')`

Output: 2

format()

Explanation: Formats values into a string.

Example: `'Hi {}'.format('Sam')`

Output: Hi Sam

format_map()

Explanation: Formats using a mapping/dictionary.

Example: `'{x}'.format_map({'x':1})`

Output: 1

index()

Explanation: Returns first index or raises ValueError.

Example: `'hello'.index('e')`

Output: 1

isalnum()

Explanation: True if all characters are letters or digits.

Example: `'abc123'.isalnum()`

Output: True

isalpha()

Explanation: True if all characters are letters.

Example: `'abc'.isalpha()`

Output: True

isascii()

Explanation: True if all characters are ASCII.

Example: `'abc'.isascii()`

Output: True

isdecimal()

Explanation: True if all are decimal digits.

Example: `'123'.isdecimal()`

Output: True

isdigit()

Explanation: True if all are digits.

Example: `'123'.isdigit()`

Output: True

isidentifier()

Explanation: Checks valid Python identifier.

Example: `'var_1'.isidentifier()`

Output: True

islower()

Explanation: Checks all letters are lowercase.

Example: `'abc'.islower()`

Output: True

isnumeric()

Explanation: Checks numeric characters.

Example: `'123'.isnumeric()`

Output: True

isprintable()

Explanation: Checks printable characters.

Example: `'abc'.isprintable()`

Output: True

isspace()

Explanation: Checks whitespace only.

Example: `' '.isspace()`

Output: True

istitle()

Explanation: Checks title case.

Example: `'Hello World'.istitle()`

Output: True

isupper()

Explanation: Checks uppercase.

Example: `'ABC'.isupper()`

Output: True

join()

Explanation: Joins iterable using separator.

Example: `'-'.join(['a', 'b'])`

Output: a-b

ljust()

Explanation: Left-aligns string.

Example: `'hi'.ljust(5, '-')`

Output: `hi---`

lower()

Explanation: Converts to lowercase.

Example: `'ABC'.lower()`

Output: `abc`

lstrip()

Explanation: Removes leading spaces.

Example: `' hi'.lstrip()`

Output: `hi`

maketrans()

Explanation: Creates translation table.

Example: `str.maketrans('a', 'b')`

Output: `translation table`

partition()

Explanation: Splits at first separator.

Example: `'a=b'.partition('=')`

Output: `('a', '=', 'b')`

removeprefix()

Explanation: Removes matching prefix.

Example: `'unhappy'.removeprefix('un')`

Output: `happy`

removesuffix()

Explanation: Removes matching suffix.

Example: `'test.py'.removesuffix('.py')`

Output: `test`

replace()

Explanation: Replaces substring.

Example: `'cat'.replace('c', 'b')`

Output: `bat`

rfind()

Explanation: Finds last occurrence.

Example: `'hello'.rfind('l')`

Output: 3

rindex()

Explanation: Last index or error.

Example: `'hello'.rindex('l')`

Output: 3

rjust()

Explanation: Right-aligns string.

Example: `'hi'.rjust(5, '-')`

Output: ---hi

rpartition()

Explanation: Splits at last separator.

Example: `'a=b'.rpartition('=')`

Output: ('a', '=', 'b')

rsplit()

Explanation: Splits from right.

Example: `'a,b,c'.rsplit(',', 1)`

Output: ['a,b', 'c']

rstrip()

Explanation: Removes trailing spaces.

Example: `'hi '.rstrip()`

Output: hi

split()

Explanation: Splits into list.

Example: `'a b'.split()`

Output: ['a', 'b']

splitlines()

Explanation: Splits at line breaks.

Example: `'a\nb'.splitlines()`

Output: ['a', 'b']

startswith()

Explanation: Checks starting prefix.

Example: `'abc'.startswith('a')`

Output: True

strip()

Explanation: Removes leading/trailing spaces.

Example: `' hi '.strip()`

Output: hi

swapcase()

Explanation: Swaps upper/lower case.

Example: `'AbC'.swapcase()`

Output: aBc

title()

Explanation: Capitalizes each word.

Example: `'hello world'.title()`

Output: Hello World

translate()

Explanation: Translates characters.

Example: `'abc'.translate(str.maketrans('a','x'))`

Output: xbc

upper()

Explanation: Converts to uppercase.

Example: `'abc'.upper()`

Output: ABC

zfill()

Explanation: Pads with leading zeros.

Example: `'42'.zfill(5)`

Output: 00042